

Lost and found portal– Web based Item recovery System

(Project Report for Copyright Submission)

Computer Software – Copyright Submission

ABSTRACT

Lost & Found Portal is a web-based platform developed to solve the problem of managing lost and found items in campuses and cities. Traditional methods such as notice boards and social media posts are unorganized and inefficient. This system provides a centralized platform where users can report lost items, post found items, and search items easily.

The platform supports image upload, category-based filtering, and messaging features to improve identification and communication. It also includes an admin approval system to verify posts and reduce false claims. Status tracking such as Open, Claimed, and Closed helps monitor item progress.

Overall, the system improves recovery rates, reduces manual effort, and provides a secure and efficient solution for lost item management.

INTRODUCTION

In campuses and cities, people frequently lose personal belongings such as wallets, mobile phones, ID cards, and other valuable items. Currently, these items are managed through informal methods like notice boards, social media, or word of mouth, which often lead to confusion, delays, and low recovery rates.

There is no structured system to track lost and found items, making it difficult for users to find their belongings or verify ownership. This creates inefficiency and frustration among users.

The Lost & Found Portal is designed to address these challenges by providing a centralized and digital solution. The system allows users to report lost items, post found items, and communicate with each other through a single platform. With features like image upload, filtering, messaging, and admin control, the system ensures better organization, improved communication, and higher chances of item recovery.

OBJECTIVES

- To create a centralized platform for managing lost and found items.
 - To allow users to post lost and found items easily.
 - To provide image upload for better identification.
 - To implement a messaging system for communication.
 - To ensure data security through authentication.
 - To improve item recovery rate.
 - To provide admin control for system management.

SYSTEM OVERVIEW

Lost & Found Portal is developed as a full-stack web application using the MERN (MongoDB, Express.js, React, Node.js) architecture. The system is divided into two main parts: the Public Site and the Management Site.

The frontend is built using React, providing a user-friendly and responsive interface. Users can register, log in, post items, upload images, and search for items easily.

The backend is developed using Node.js and Express.js, which handles API requests, business logic, and communication with the database. MongoDB is used to store user data, item details, categories, and messages.

The system follows a client-server architecture where the frontend communicates with the backend using REST APIs. Admin approval and status tracking ensure proper management of items.

FEATURES

- **Item Posting:**

Users can post both lost and found items by entering important details such as item name, description, category, location, and date. This structured format helps in organizing all item information properly. It replaces unorganized methods like notice boards and ensures that every item is recorded clearly, making it easier for others to search and identify.

- **Image Upload:**

This feature allows users to upload images of the item while posting. Images provide a clear visual representation, which makes identification faster and more accurate. It reduces confusion caused by text-only descriptions and increases the chances of returning the item to the correct owner.

- **Category & Location Filtering:**

Users can filter items based on categories (such as electronics, documents, accessories) and locations where the item was lost or found. This helps users quickly find relevant items without checking all posts. It improves search efficiency, saves time, and makes the system more user-friendly, especially when many items are listed.

- **Messaging System:**

The messaging system allows users to communicate directly with each other within the platform. If a user finds a matching item, they can contact the person who posted it. This avoids sharing personal contact details publicly and ensures safe and quick communication, helping in faster recovery of items.

- **Item Status Tracking (Open / Claimed / Closed):**

Each item has a status to indicate its current condition. “Open” means the item is still lost or available. “Claimed” shows that someone has requested or identified it. “Closed” means the item has been successfully returned. This feature keeps the system organized and helps users understand the progress of each item.

- **User Authentication:**

Users must register and log in to access the system. Authentication is implemented using JWT (JSON Web Token), which ensures secure access. It protects user data, prevents unauthorized usage, and allows only valid users to post or interact with items, making the system reliable and secure.

- **Admin Approval:**

Before any item is visible on the public platform, it must be approved by an admin. The admin checks whether the post is genuine and complete. This helps in preventing fake posts, spam, and misuse of the system. It also increases trust among users by ensuring that all displayed items are verified.

- **Dashboard:**

The dashboard provides a summary view of the system. Users can see their posted items, item status, and activity history. Admins can monitor total users, items, and pending approvals. It helps in easy management, quick access to important data, and better control over the system.

TECHNOLOGY STACK

Backend:

Node.js is used to run JavaScript on the server side. It handles user requests, processes data, and manages the backend logic of the application. It is fast and efficient, which makes the system scalable and capable of handling multiple users at the same time.

Express.js is a framework built on Node.js that is used to create APIs easily. It helps in handling routes, managing requests and responses, and organizing the backend code. It makes the development process simple and structured.

MongoDB is used to store all application data such as user details, item information, categories, and messages. It stores data in a flexible JSON-like format, which makes it easy to manage and update. It is suitable for applications with dynamic data.

Mongoose is used to define schemas and models for MongoDB. It helps in structuring the data, validating inputs, and interacting with the database in a proper way. It makes database operations easier and more organized.

JWT (JSON Web Token) is used for secure user authentication. When a user logs in, a token is generated and sent to the client. This token is used to verify the user for future requests, ensuring that only authorized users can access the system.

bcrypt is used to encrypt user passwords before storing them in the database. It converts passwords into a hashed format, which improves security and protects user data from unauthorized access.

Frontend:

React.js is used to build the user interface of the application. It allows developers to create reusable components and provides a fast and interactive user experience. It also helps in managing the application state efficiently.

React Router is used to handle navigation between different pages in the application. It allows users to move between pages like login, dashboard, and item list without reloading the page, making the application smooth and user-friendly.

Axios is used to send HTTP requests from the frontend to the backend. It helps in fetching data from the server, sending user inputs, and handling responses efficiently.

CSS or Tailwind CSS is used for designing the user interface. It helps in creating a clean, responsive, and modern layout. Tailwind CSS provides ready-to-use utility classes that speed up the design process.

SYSTEM ARCHITECTURE

The overall working of the system follows a step-by-step flow:

- The user performs an action on the frontend, such as logging in, posting an item, or searching for an item.
- The frontend sends an API request to the backend server.
- The backend receives the request, processes the logic, and interacts with the database if needed.
- The database stores or retrieves the required data.
- The backend sends a response back to the frontend.
- The frontend updates the user interface based on the response received.

This workflow ensures smooth operation, fast response time, and efficient data handling across the entire system.

MODULE DESCRIPTION

Authentication Module:

This module manages user registration and login functionality. Users are required to create an account by providing basic details such as name, email, and password. The passwords are securely hashed using bcrypt before being stored in the database to ensure security. Upon successful login, a JSON Web Token (JWT) is generated and sent to the user, which is used to authenticate protected routes. This ensures that only authorized users can access system features like posting items and messaging.

Item Management Module:

This module handles all operations related to lost and found items. It allows users to post new items by providing details such as title, description, category, location, and images. Users can also update or delete their posts if needed. The module ensures that all item data is properly stored and managed in the database. It also works with the admin module for approval and status updates of items.

Category Module:

This module is responsible for organizing items into different categories such as electronics, documents, accessories, and others. It helps users easily filter and search for items based on their type. Admins can add, update, or delete categories as required. This improves the overall usability and efficiency of the system.

Messaging Module:

This module enables communication between users within the platform. When a user finds a matching item, they can send a message to the person who posted it. This helps users exchange necessary information securely without sharing personal contact details publicly. The module stores all messages and ensures smooth and safe interaction between users.

Admin Module:

This module provides control and management features for administrators. Admins can view and manage user data, approve or reject item posts, and monitor system activities. It ensures that only verified and appropriate content is displayed on the platform. The admin module helps maintain system security, reliability, and proper functioning.

DATABASE DESIGN

The Lost & Found Portal application uses MongoDB as its primary database to store and manage all application data. Mongoose is used as an Object Data Modeling (ODM) library to define schemas and handle database operations efficiently. The database is structured using multiple models, each representing a specific component of the system.

User Model:

This model stores user-related information such as name, email, and hashed password. It also includes additional details like user contact information and activity history. The model ensures secure password storage using encryption and supports authentication processes such as login and access control.

Item Model:

This model defines the structure of lost and found items in the system. It includes fields such as item title, description, category, location, date, image, and status (Open, Claimed, Closed). Each item is linked to a specific user who posted it. This model helps in organizing and managing item data efficiently.

Category Model:

This model stores different item categories such as electronics, documents, accessories, and others. It helps in classifying items and allows users to filter and search items easily. Admins can manage categories by adding, updating, or deleting them.

Message Model:

This model stores communication between users. It includes fields such as sender ID, receiver ID, message content, and timestamp. This helps users communicate securely within the platform regarding lost or found items.

The database design ensures efficient data storage, easy retrieval, and scalability, supporting smooth operation and future expansion of the system.

WORKING OF SYSTEM

1. **User Registration and Login:**

New users register by providing basic details such as name, email, and password. The password is securely hashed before being stored in the database. Existing users can log in using their credentials and receive a JSON Web Token (JWT) for authenticated access to the system.

2. **Dashboard Access:**

After successful login, users are directed to the dashboard, which displays key information such as their posted items, item status, and recent activity. The dashboard provides a quick overview and easy navigation to different features of the system.

3. **Item Posting:**

Users can post lost or found items by entering details such as item name, description, category, location, date, and uploading an image. This structured information helps in proper identification and organization of items within the system.

4. **Admin Approval:**

Once an item is posted, it is sent to the admin for verification. The admin reviews the details and approves or rejects the post. Only approved items are made visible on the public platform, ensuring authenticity and preventing misuse.

5. **Item Display and Search:**

Approved items are displayed on the public platform. Users can browse or search for items using filters such as category and location. This helps users quickly find relevant items without going through all available posts.

6. **Messaging and Communication:**

If a user finds a matching item, they can contact the owner through the messaging system. This allows users to communicate securely within the platform and exchange necessary details for item recovery.

7. **Item Claim and Status Update:**

When an item is successfully identified, the user can claim it. The item status is updated from “Open” to “Claimed” and finally to “Closed” once the item is returned. This helps in tracking the progress of each item.

8. **Data Storage and Tracking:**

All user actions such as item posts, messages, and status updates are stored in the database. This allows the system to maintain records, track activity, and provide users with updated information at all times.

RESULTS

- The Lost & Found Portal system has been successfully developed and tested, demonstrating reliable performance across all core functionalities. The system effectively allows users to post lost and found items with detailed information and images. The item management process works smoothly, ensuring proper organization and easy accessibility of data.
- The item listing and filtering features function efficiently, allowing users to search items based on category and location. This improves usability and helps users quickly find relevant items without confusion or delay.
- The messaging system enables smooth and secure communication between users. It allows users to contact item owners directly within the platform, which improves coordination and increases the chances of successful item recovery.
- Overall, the system provides a responsive and user-friendly experience with efficient data handling and secure authentication. The Lost & Found Portal effectively solves the problem of lost item management by offering a practical, organized, and reliable solution.

CONCLUSION

- The Lost & Found Portal successfully addresses the common problem of managing lost and found items in campuses and urban areas. By replacing manual methods with a digital platform, the system provides an efficient and organized solution for reporting, searching, and recovering lost items.
- The platform ensures smooth communication between users through its messaging system and improves item identification using image uploads and structured data. Features like admin approval and status tracking enhance reliability and transparency, helping users trust the system and increasing the chances of successful item recovery.
- Built using modern web technologies, the system is scalable, secure, and easy to maintain. The Lost & Found Portal demonstrates how technology can be effectively used to solve real-life problems, making it a practical and valuable solution for item management in real-world environments.

FUTURE SCOPE

- The Lost & Found Portal can be enhanced by integrating location-based services such as Google Maps, allowing users to tag and view the exact location where an item was lost or found. This will improve accuracy and make it easier for users to locate items.
- The system can be extended into a mobile application for Android and iOS platforms, providing better accessibility and convenience for users to report and search items anytime and anywhere.
- Advanced technologies such as Artificial Intelligence can be implemented to enable smart item matching. The system can automatically suggest possible matches between lost and found items based on images, category, and description.
- A notification system can be added to provide real-time updates to users about item status changes, new matching items, or messages. This will improve user engagement and ensure timely communication.
- Multi-language support can be introduced to make the platform accessible to a wider audience, allowing users from different regions to use the system comfortably in their preferred language.

DECLARATION

This project titled “**Lost & Found Portal – Web-Based Item Recovery System**” is an original work developed by the authors. The entire source code, design, and documentation have been created independently without any unauthorized use of existing copyrighted material.

All the concepts, implementation, and content presented in this project are the result of our own efforts and learning. This project is submitted for copyright registration as a genuine and original software work.